

CERTIFICATE OF AUTHORSHIP

Instructions

- Download and fill this PDF form completely.
- Each course requirement submission, unless otherwise specified by the Course Instructor, whether in electronic or paper form, must be accompanied by a corresponding properly accomplished Certificate of Authorship.

Description of Submission

Title of Submission: Midterm Project: Guitar

Type of Submission: ☐ Program ☒ Project ☐ Report ☐ Paper
☐ Other (specify) _____

Date of Submission: September 27, 2025

Certification

I hereby certify that the submission described in this document abides by the principles stipulated in the DISCS Academic Integrity Policy document. I further certify that I am the author of this submission and that any assistance I received in its preparation is fully acknowledged and disclosed in the documentation. I have also cited all sources from which I obtained data, ideas, or words that are directly copied or paraphrased in this document. Sources are properly credited according to accepted standards for professional publication.

Karplus-Strong Algorithm (basis of the entire project):
https://en.wikipedia.org/wiki/Karplus%E2%80%93Strong_string_synthesis

(Extra) Decay and Sustain Concept (ADSR):
[https://en.wikipedia.org/wiki/Envelope_%28music%29?](https://en.wikipedia.org/wiki/Envelope_%28music%29?text=The%20first%20way%20is%20plugging%20other%20coordinate%20for%20that%20value)

(Extra) Linear Interpolation (lerp) Formula for Decay by Frequency:
<https://www.indeed.com/career-advice/career-development/how-to-interpolate-in-excel#:~:text=The%20first%20way%20is%20plugging,other%20coordinate%20for%20that%20value>.

Python Dictionary (for pairing keys and strings together): https://www.w3schools.com/python/python_dictionaries.asp

Number Clamping/Clipping Techniques (during total sample calculation, for safety):
<https://stackoverflow.com/questions/4092528/how-can-i-clamp-clip-restrict-a-number-to-some-range>

Declaration of Use of Generative AI

Tool: _____

Purpose: _____

I have reviewed and revised the content as I see fit. I take full responsibility for the content and ownership of the submitted / published work.

Student and Course Information

Student's Full Name: Ezekiel Villasurda

Course Code & Section: CSCI 30 - L1

Course Title: DATA STRUCTURES AND ALGORITHMS

Course Instructor: Brian Christopher Guadalupe

Signature: _____

